



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

May 29, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 8299
(TWENTY-THIRD REVISION)

EXPIRATION DATE: 2029-04-30

(FOR RENEWAL, SEE 49 CFR § 107.109)

1. GRANTEE: Meggitt Safety Systems, Inc.
Simi Valley, CA
2. PURPOSE AND LIMITATIONS:
 - a. This special permit authorizes the manufacture, mark, sale and use of a non-DOT specification cylinder conforming to all regulations applicable to a DOT Specification 3HT, except as specified herein, for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171- 180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.304(a) in that a non-DOT specification cylinder is not authorized, except as specified herein.
5. BASIS: This special permit is based on Meggitt Safety Systems Inc.'s application dated February 5, 2025, submitted in accordance with § 107.109.

Tracking Number: 2025024170

6. HAZARDOUS MATERIALS (49 CFR § 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Helium, compressed	2.2	UN1046	N/A
Nitrogen, compressed	2.2	UN1066	N/A

7. SAFETY CONTROL MEASURES: Prescribed packaging is a non-DOT specification welded stainless steel cylinder manufactured in compliance with HTL drawings on file with the Office of Hazardous Materials Safety and with DOT Specification 3HT (§ 178.44), except as follows:

§ 178.44(a) Type, size and service pressure.

Welded spheres (two seamless hemispheres) or circumferentially welded cylinders (2 seamless drawn shells) with a water capacity (nominal) of not over 400 cubic inches and a maximum service pressure of 3,300 pounds per square inch.

§ 178.44(b) Authorized steel.

Type ARMCO Nitric 40 (21-6-9) stainless steel with the following analysis is authorized with proper welding procedure.

CHEMICAL ANALYSIS

	<u>PERCENT</u>
Carbon	0.040 max.
Manganese	8.00/10.00
Phosphorus	0.060 max.
Sulfur	0.030 max.
Silicon	1.00 Max.
Chromium	19.00/21.50
Nickel	5.50/7.50
Nitrogen	0.15/0.40

§ 178.44(d) Manufacture.

* * *

Each cylinder must be subjected to a process treatment after welding and before stress relieving. Process treatment must consist of hydrostatic pressurization at a pressure (maintained for at least 3 minutes) of at least 100% but not more than 110% of prescribed test pressure. The process treatment must be witnessed by the independent inspector.

§ 178.44(e) Welding or Brazing.

Welding, as prescribed in § 178.44(a) of this special permit, is authorized. All pressure welds must be 100 percent radiographed after the hydrostatic test.

§ 178.44(f) Wall thickness.

(1) Applies except that 112,500 psi is substituted for 105,000 psi at the end of the section.

(2) and (3) * * *

(4) Calculations for spheres must be made by the formula:

$$S = PD/4tE$$

Where:

E = 0.85 which must be applied to the girth weld area and heat zones. The zones must extend a distance of 6 times the wall thickness from the center of the weld.

E = 1.0 (for all other areas).

§ 178.44(g) Heat treatment.

The hemispheres and half cylinders may be stress relieved or annealed for forming. Welded cylinders must be stress relieved at a temperature of $900^{\circ}\text{F} \pm 25^{\circ}\text{F}$ for one hour. Stress relieving is to be done after process treatment and before the hydrostatic test.

§ 178.44(j) Cycling tests.

Applies except that cycle test on each lot is not required.

§ 178.44(l) Flattening test.

A flattening test must be performed on one sphere, taken at random out of each lot of 200 or less after the hydrostatic test, by placing the sphere between parallel plates on a press with the welded seam at right angles to the plate.

§ 178.44(m) Physical tests.

* * *

Physical test is not required on spheres or cylinders having a water capacity of 60 cubic inches or less.

(1) * * *

(2) * * *

(i) Applies except for substituting a gauge length of 2 inches.

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§ 178.44(n) Magnetic particle inspection.

Not required. Instead, each cylinder must be inspected using apparatus and procedures for liquid penetrant examination in accordance with ASTM E-165-95. Inspection must be performed externally on the finished cylinder after the hydrostatic test. Evidence of discontinuities, which in the opinion of the independent inspector may appreciably weaken or decrease the durability of the cylinder, will be cause of rejection.

§ 178.44(p) Acceptable results of tests.

* * *

(1) Flattening required without cracking to ten times the wall thickness of the cylinder or to 50 percent of the original outside diameter for a sphere.

(2) Physical tests:

(i) Elongation at least 6 percent in gauge length of 24 times wall thickness, and at least 10 percent in gauge length of 2 inches.

(ii) Tensile strength must be at least 150,000 psi.

(3) Burst pressure must be at least 4/3 times the test pressure. Actual burst pressure must be recorded.

(4) Cycling: The design must be qualified by cycling three cylinders from zero to service pressure at a rate not to exceed ten cycles per minute to at least 50,000 pressurizations without evidence of distortion or failure. Each design change must be qualified by cycling at least one cylinder at the above rate and acceptance criteria.

§ 178.44(q) Rejected cylinders.

Repair of welded seams by welding prior to process treatment authorized; subsequent thereto cylinders must be heat treated and pass all prescribed tests.

§ 178.44(r) Marking.

Applies except that instead of DOT-3HT, cylinders must be marked “DOT-SP 8299” followed by the service pressure. Rejection elastic expansion is not required for cylinders having a water capacity of 60 cubic inches or less.

8. SPECIAL PROVISIONS:

b. Cylinders must be shipped in strong outside packagings in conformance with § 173.301(a)(9).

c. Cylinders manufactured under this special permit are not authorized for use fifteen (15) years after the date of manufacture except cylinders which were manufactured under P/N 36200021-1 or P/N 36200021-2 are authorized for use for up to twenty-four (24) years from the date of manufacture.

d. Each cylinder must be reinspected and hydrostatically retested every 5 years in accordance with § 180.209 as prescribed for DOT-3HT cylinders.

e. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special

permit.

f. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.

g. A current copy of this special permit must be maintained at each facility where the package is offered for transportation.

h. Each cylinder manufactured under the authority of this special permit must be marked with the registration symbol designated by the Office of Hazardous Materials Safety for a specific manufacturing facility.

i. A current copy of this special permit, in its current status, must be maintained at each manufacturing facility at which this packaging is manufactured and must be made available to a DOT representative upon request.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo-only aircraft and passenger-carrying aircraft.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
 - o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when the special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) — “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: BB